Minqi's Solutions to the Exercises of Walter Rudin (1976), Principles of Mathematical Analysis (3rd Edition)

Minqi Pan[1]

[1]Website: http://www.minqi-pan.com

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Preface

Remarkable exercises are:

(1) Exercise 2.6: E, E' and $\bar{E}$;
(2) Exercise 3.3: $s_{n+1} = \sqrt{2 + \sqrt{s_n}}$;
(3) Exercise 3.5: Upper Limits and Algebraic Operations.

Minqi Pan
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CHAPTER 1

The Real and Complex Number Systems

1.1. Exercise 1: Mixed Operations between Rational and Irrational Numbers Produce Irrational Numbers

If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx are irrational.

PROOF. Denote $\alpha = r + x, \beta = rx$.

Suppose $\alpha \in \mathbb{Q}$. Then $x = \alpha - r$ is also in $\mathbb{Q}$, because $\mathbb{Q}$ is a field that is closed with respect to additive inverse and addition. This contradicts with the presumption that $x \notin \mathbb{Q}$.

Suppose $\beta \in \mathbb{Q}$. Then $x = \beta/r$ is also in $\mathbb{Q}$, because $r \neq 0$ and $\mathbb{Q}$ is a field that is closed with respect to multiplicative inverse and multiplication. This contradicts with the presumption that $x \notin \mathbb{Q}$. □

1.2. Exercise 2: $\sqrt{12}$ is an Irrational Number

Prove that there is no rational number whose square is 12.

LEMMA 1.1. $\forall n \in \mathbb{Z}, \forall p \in \{2, 3\}$,

$$n^2 \equiv 0 \implies n \equiv 0 \pmod{p}.$$

PROOF. We check the squares of every element of $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$, the ring of integers modulo 2.

$$0^2 = 0$$
$$1^2 = 1$$

Thus $n^2 = 0$ implies $n = 0$ in $\mathbb{Z}/2\mathbb{Z}$ by exhaustion.

Similarly we check the squares of every element in $\mathbb{Z}/3\mathbb{Z} = \{0, 1, 2\}$, the ring of integers modulo 3.

$$0^2 = 0$$
$$1^2 = 1$$
$$2^2 = 1$$

Thus $n^2 = 0$ implies $n = 0$ in $\mathbb{Z}/3\mathbb{Z}$ by exhaustion. □

LEMMA 1.2. *There is no rational number whose square is* 3.

PROOF. We now show that the equation

$$p^2 = 3 \tag{1.1}$$

is not satisfied by any rational p. If there was such a p, we could write $p = \frac{m}{n}$ where $m \equiv 0 \pmod{3}$ and $n \equiv 0 \pmod{3}$ are not both true. Let us assume this is done. Then (1.1) implies

$$m^2 = 3n^2, \tag{1.2}$$

This shows that $m^2 \equiv 0 \pmod{3}$ and we have $m \equiv 0 \pmod{3}$ by Lemma 1.1. So $m^2 \equiv 0 \pmod{9}$. It follows that the right side of (1.2) is divisible by 9, so that $n^2 \equiv 0 \pmod{3}$, which implies that $n \equiv 0 \pmod{3}$ again by Lemma 1.1.

The assumption that (1.1) holds thus leads to the conclusion that both $m \equiv 0 \pmod 3$ and $n \equiv 0 \pmod 3$ are true, contrary to our choice of m and n. Hence (1.1) is impossible for rational p. $\square$

Main Proof. Suppose that there existed $p \in \mathbb{Q}$ such that $p^2 = 12$. Then there exists $m, n \in \mathbb{Z}$ such that

$$p = \frac{m}{n} \implies 12 = \frac{m^2}{n^2} \implies m^2 = 2 \times 6n^2 \implies m^2 \equiv 0 \pmod 2.$$

So it follows from Lemma 1.1 that

$$m \equiv 0 \pmod 2.$$

Therefore there exists $k \in \mathbb{Z}$ such that $m = 2k$, and

$$4k^2 = 12n^2 \implies k^2 = 3n^2 \implies \left(\frac{k}{n}\right)^2 = 3.$$

This contradicts with Lemma 1.2. $\square$

CHAPTER 2

Basic Topology

2.1. Exercise 1: Empty Set is a Subset of Every Set

Prove that the empty set is a subset of every set.

PROOF. Given a set X. It is true that $x \in \varnothing$ implies any statement about X, because $x \in \varnothing$ is always false.

In particular, $x \in \varnothing$ implies $x \in X$. Thus $\varnothing \subset X$. $\square$

2.2. Exercise 2: Algebraic Numbers are Countable

A complex number z is said to be algebraic if there are integers $a_0, \dots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n = 0.$$

Prove that the set of all algebraic numbers is countable.

PROOF. Denote $\mathcal{A}$ as the set of all algebraic numbers.

For each $N \in \{2, 3, 4, \dots\}$, define $\mathcal{A}_N$ to be the set of all $z \in \mathcal{A}$ which satisfy the following property: there exist $n \in \{1, 2, 3, \dots\}$ and $a_0, \dots, a_n \in \mathbb{Z}$ such that

$$a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n = 0$$

and

$$n + |a_0| + |a_1| + \cdots + |a_n| = N.$$

We claim that

$$\mathcal{A} = \bigcup_{N=2}^{\infty} \mathcal{A}_N.$$

To prove this, note that each $\mathcal{A}_N \subset \mathcal{A}$, it only suffices to show that $\mathcal{A} \subset \cup_{N=2}^{\infty} \mathcal{A}_N$. Pick $z \in \mathcal{A}$. Then, by definition, there there exist $n \in \{1, 2, 3, \dots\}$ and $a_0, \dots, a_n \in \mathbb{Z}$ such that $a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n = 0$. Put $n_0 = n + |a_0| + |a_1| + \cdots + |a_n|$, then $n_0 \geqslant 2$ and $z \in \mathcal{A}_{n_0}$. Thus $z \in \cup_{N=2}^{\infty} \mathcal{A}_N$.

For each $N \in \{2, 3, 4, \dots\}$, observe that there are only finite choices of

$$n, a_0, a_1, \dots, a_n$$

such that $n + |a_0| + |a_1| + \cdots + |a_n| = N$. Further, for each choice of $n, a_0, a_1, \dots, a_n$, the equation $a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n = 0$ has exactly n solutions of $z \in \mathbb{C}$ as shown by 4.14 of [**Axl14**]. Thus each $\mathcal{A}_N$ is a finite set. By the corollary of Theorem 2.12 of [**Rud76**], $\mathcal{A}$ is at most countable.

Finally, since $\mathbb{Q} \subset \mathcal{A}$, $\mathcal{A}$ is infinite. Therefore $\mathcal{A}$ is countable. $\square$

2.3. Exercise 3: There Exist Non-Algebraic Real Numbers

Prove that there exist real numbers which are not algebraic.

PROOF. Suppose all real numbers are algebraic. Then $\mathbb{R}$ would be countable as shown in Exercise 2. This contradicts with the corollary of Theorem 2.43 of [**Rud76**]. □

2.4. Exercise 4: Irrational Real Numbers are Uncountable

Is the set of all irrational real numbers countable?

PROOF. We claim that the set of all irrational real numbers is uncountable.

By the corollary of Theorem 2.13 of [**Rud76**], $\mathbb{Q}$ is countable. Note that $\mathbb{R}$ is the union of $\mathbb{Q}$ and the set of all irrational real numbers. If the set of all irrational real numbers was countable, then, by the corollary of Theorem 2.12 of [**Rud76**], $\mathbb{R}$ would be countable, which would contradict with Theorem 2.43 of [**Rud76**]. □

2.5. Exercise 5: A Real Bounded Set with 3 Limit Points

Construct a bounded set of real numbers with exactly three limit points.

PROOF. Let

$$\begin{aligned}
E_0 &= \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}, \\
E_1 &= \left\{1+\frac{1}{2}, 1+\frac{1}{3}, 1+\frac{1}{4}, \dots\right\}, \\
E_2 &= \left\{2+\frac{1}{2}, 2+\frac{1}{3}, 2+\frac{1}{4}, \dots\right\}, \\
E &= E_0 \cup E_1 \cup E_2.
\end{aligned}$$

We claim that E is a bounded set of real numbers with exactly three limit points: $0, 1, 2$.

Set $M = 3$ and $q = 0$, then $d(p, q) < M$ for all $p \in E$. Thus E is bounded.

Pick $p \in \{0, 1, 2\}$ and a neighborhood $N_r(p)$ of p. Let

$$q = p + \frac{1}{\lceil \frac{1}{r} \rceil + 1},$$

then $q \neq p$, $q \in N_r(p)$ and $q \in E$.

Pick $p \in \mathbb{R}$ such that $p \notin \{0, 1, 2\}$, then there exists a neighborhood of p that contains only finitely many points of E. Thus p is not a limit point of E by Theorem 2.20 of [**Rud76**]. □

2.6. Exercise 6: E, E' and $\bar{E}$

Let E' be the set of all limit points of a set E. Let $\bar{E} = E \cup E'$.

(1) Prove that E' is closed.
(2) Prove that E and $\bar{E}$ have the same limit points.
(3) Do E and E' always have the same limit points?

PROOF. (1) Suppose p is a limit point of E'. Pick a neighborhood $N_r(p)$ of p. Then there exists $q \in N_r(p)$ such that $q \neq p$ and $q \in E'$. Let

$$t = \min\left\{\frac{d(p,q)}{2}, \frac{r - d(p,q)}{2}\right\},$$

then there exists $s \in N_t(q)$ such that $s \neq q$ and $s \in E$. Consequently,

$$\begin{aligned} d(s,p) &\leqslant d(s,q) + d(q,p) \\ &< t + d(q,p) \\ &\leqslant \frac{r - d(p,q)}{2} + d(q,p) \\ &= \frac{d(q,p) + r}{2} \\ &< \frac{r + r}{2} \\ &= r. \end{aligned}$$

Hence $s \in N_r(p)$. Moreover,

$$\begin{aligned} d(s,p) &\geqslant d(p,q) - d(q,s) \\ &> d(p,q) - t \\ &\geqslant d(p,q) - \frac{d(p,q)}{2} \\ &= \frac{d(p,q)}{2} \\ &> 0. \end{aligned}$$

Hnece $s \neq p$. Therefore p is also a limit point of E, which implies $p \in E'$.

We have shown above that every limit point of E' is in E'. Thus E' is closed.

(2) Suppose p is a limit point of E. Pick a neighborhood $N_r(p)$. Then there exists $q \in N_r(p)$ such that $q \neq p$ and $q \in E$. Since $E \subset \bar{E}$, it follows that $q \in \bar{E}$. Hence p is also a limit point of $\bar{E}$.

On the other hand, suppose p is a limit point of $\bar{E}$. Pick a neighborhood $N_r(p)$. Then there exists $q \in N_r(p)$ such that $q \neq p$ and $q \in \bar{E}$. If $q \in E$, let $s = q$. If $q \in E'$, then obtain s using the same notation as (1). Either way, there exists $s \in N_r(p)$ such that $s \in E$ and $s \neq p$. Hence p is also a limit point of E.

(3) No, E and E' do not always have the same limit points. Suppose

$$E = \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\}.$$

Then

$$\begin{aligned} E' &= \{0\}, \\ E'' &= \varnothing. \end{aligned}$$

However, in (1), we have proven that $E'' \subset E'$. Hence, in general, E has more limit points than E'.

□

CHAPTER 3

Numerical Sequences and Series

3.1. Exercise 1: Convergence of $\{s_n\}$ Implies Convergence of $\{|s_n|\}$

Prove that convergence of $\{s_n\}$ implies convergence of $\{|s_n|\}$. Is the converse true?

PROOF. Suppose

$$s_n \to p.$$

We claim that

$$|s_n| \to |p|.$$

In order to prove this, pick $\varepsilon > 0$. Then there exists an integer N such that $n \geqslant N$ implies that $|s_n - p| < \varepsilon$. By the reverse triangle inequality,

$$||s_n| - |p|| \leqslant |s_n - p|\,,$$

Hence $n \geqslant N$ also implies that

$$||s_n| - |p|| < \varepsilon.$$

Finally, we claim that the converse is not true. Suppose

$$s_n = (-1)^n,$$

then $|s_n| \to 1$ yet s_n diverges. □

3.2. Exercise 2: $\lim_{n\to\infty}(\sqrt{n^2+n} - n) = 2$

Calculate $\lim_{n\to\infty}(\sqrt{n^2+n} - n)$.

PROOF. We claim that

$$\lim_{n\to\infty}\left(\sqrt{n^2+n} - n\right) = 2.$$

To prove this, define

$$f(x) = \sqrt{1+x} + 1.$$

Then

$$\begin{aligned}
\lim_{n\to\infty}\left(\sqrt{n^2+n} - n\right) &= \lim_{n\to\infty}\left(\frac{\sqrt{n^2+n}+n}{n}\right) \\
&= \lim_{n\to\infty}\left(\sqrt{\frac{n^2+n}{n^2}} + 1\right) \\
&= \lim_{n\to\infty}\left(\sqrt{1+\frac{1}{n}} + 1\right) \\
&= \lim_{n\to\infty} f(\frac{1}{n}).
\end{aligned}$$

By Theorem 3.20 of [**Rud76**],

$$\lim_{n\to\infty}\frac{1}{n} = 0.$$

Since $f(x)$ is continuous on a neighborhood $N_r(0)$ of $\mathbb{R}$, Theorem 4.6 of [**Rud76**] implies that

$$\lim_{x\to 0} f(x) = f(0).$$

Finally, by Theorem 4.2 of [**Rud76**],

$$\lim_{n\to\infty} f(\frac{1}{n}) = \lim_{x\to 0} f(x) = f(0) = 2.$$

□

3.3. Exercise 3: $s_{n+1} = \sqrt{2+\sqrt{s_n}}$

If $s_1 = \sqrt{2}$, and

$$s_{n+1} = \sqrt{2+\sqrt{s_n}} \quad (n = 1, 2, 3, \dots),$$

prove that $\{s_n\}$ converges, and that $s_n < 2$ for $n = 1, 2, 3, \dots$.

Proof. (1) We claim that s_n is bounded by 1 and 2, i.e.

$$1 < s_n < 2.$$

We prove this by induction. Clearly $1 < s_1 < 2$. Suppose $1 < s_n < 2$, then

$$\begin{gathered}
1 < s_n < 2 \\
\Downarrow \\
\sqrt{1} < \sqrt{s_n} < \sqrt{2} \\
\Downarrow \\
2+\sqrt{1} < 2+\sqrt{s_n} < 2+\sqrt{2} \\
\Downarrow \\
1 = \sqrt{0+\sqrt{1}} < \sqrt{2+\sqrt{1}} < \sqrt{2+\sqrt{s_n}} < \sqrt{2+\sqrt{2}} < \sqrt{2+2} = 2 \\
\Downarrow \\
1 < s_{n+1} < 2.
\end{gathered}$$

(2) Further, we claim that s_n is strictly monotonically increasing, i.e.

$$s_n < s_{n+1}.$$

We also prove this by induction. First of all,

$$\begin{gathered}
0 < \sqrt{\sqrt{2}} \\
\Downarrow \\
2+0 < 2+\sqrt{\sqrt{2}} \\
\Downarrow \\
\sqrt{2+0} < \sqrt{2+\sqrt{\sqrt{2}}} \\
\Downarrow \\
s_1 < s_2.
\end{gathered}$$

Suppose $s_n < s_{n+1}$, then

$$\begin{gathered} s_n < s_{n+1} \\ \Downarrow \\ \sqrt{s_n} < \sqrt{s_{n+1}} \\ \Downarrow \\ 2+\sqrt{s_n} < 2+\sqrt{s_{n+1}} \\ \Downarrow \\ \sqrt{2+\sqrt{s_n}} < \sqrt{2+\sqrt{s_{n+1}}} \\ \Downarrow \\ s_{n+1} < s_{n+2}. \end{gathered}$$

In light of (1) and (2) above, Theorem 3.14 of [**Rud76**] implies that $\{s_n\}$ converges. □

3.4. Exercise 4: Upper and Lower Limits of an Iterative Sequence

Find the upper and lower limits of the sequence $\{s_n\}$ defined by

$$s_1 = 0; \quad s_{2m} = \frac{s_{2m-1}}{2}; \quad s_{2m+1} = \frac{1}{2} + s_{2m}.$$

Proof. We claim that

$$s_{2m} = \frac{1}{2} - \frac{1}{2^m}. \tag{3.1}$$

We prove this by induction. First of all, when $m = 1$,

$$s_{2m} = s_2 = \frac{s_1}{2} = 0$$

and

$$\frac{1}{2} - \frac{1}{2^m} = \frac{1}{2} - \frac{1}{2} = 0.$$

Suppose (3.1) holds for $m \geq 1$, then

$$\begin{aligned} s_{2(m+1)} &= \frac{s_{2m+1}}{2} \\ &= \frac{\frac{1}{2} + s_{2m}}{2} \\ &= \frac{1}{4} + \frac{\frac{1}{2} - \frac{1}{2^m}}{2} \\ &= \frac{1}{2} - \frac{1}{2^{m+1}}. \end{aligned}$$

Equation (3.1) further implies that

$$\begin{aligned} s_{2m+1} &= \frac{1}{2} + s_{2m} \\ &= \frac{1}{2} + \frac{1}{2} - \frac{1}{2^m} \\ &= 1 - \frac{1}{2^m}. \end{aligned} \tag{3.2}$$

Let E be the set of numbers x (in the extended real number system) such that $s_{n_k} \to x$ for some subsequence $\{s_{n_k}\}$. Then it is clear from (3.1) and (3.2) that

$$E = \{\frac{1}{2}, 1\}.$$

Therefore

$$\limsup s_n = 1, \quad \liminf s_n = \frac{1}{2}.$$

□

3.5. Exercise 5: Upper Limits and Algebraic Operations

For any two real sequences $\{a_n\}, \{b_n\}$, prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leqslant \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n,$$

provided the sum on the right is not of the form $\infty - \infty$.

LEMMA 3.1. *For any bounded real sequence* $\{a_n\}$,

$$\limsup_{n\to\infty} a_n = \lim_{k\to\infty} \left(\sup\{a_i : i \geqslant k\}\right).$$

PROOF. Since $\{a_n\}$ is bounded, the set $\{a_i : i \geqslant k\}$ is also bounded for all $k = 1, 2, 3, \ldots$. Hence

$$\sup\{a_i : i \geqslant k\}$$

exist for all $k = 1, 2, 3, \ldots$ because of the least-upper-bound property of $\mathbb{R}$.

Moreover, $\sup\{a_i : i \geqslant k\}$ is bounded below by $\sup\{a_i : i \geqslant 1\}$ and monotonically decreasing. Theorem 3.14 of [**Rud76**] thus implies that

$$\lim_{k\to\infty} \left(\sup\{a_i : i \geqslant k\}\right)$$

exists, which we denote by p.

(1) Suppose $p < \limsup_{n\to\infty} a_n$, then there exist $x \in \mathbb{R}$ and subsequence $\{a_{n_k}\}$ such that

$$p < x \leqslant \limsup_{n\to\infty} a_n$$

and

$$a_{n_k} \to x.$$

Therefore, there exists K_1 such that $k > K_1$ implies

$$\frac{p+x}{2} < a_{n_k}.$$

Moreover, there exists K_2 such that $k > K_2$ implies

$$\sup\{a_i : i \geqslant k\} < \frac{p+x}{2}.$$

Now pick $k > \max(K_1, K_2)$, then

$$\frac{p+x}{2} < a_{n_k} \leqslant \sup\{a_i : i \geqslant k\} < \frac{p+x}{2},$$

which is a contradiction.

(2) Suppose $\limsup_{n\to\infty} a_n < p$. Then we can choose K_1 such that $k > K_1$ implies

$$\sup\{a_i : i \geqslant k\} < p + \frac{1}{2}$$

and by the definition of supremum, we can choose $k_1 > K_1$ such that

$$a_{k_1} < p + \frac{1}{2}.$$

Having chosen $K_1, \ldots, K_{i-1}$, we can choose K_i such that $K_i > K_{i-1}$ and $k > K_i$ implies

$$\sup\{a_i : i \geqslant k\} < p + \frac{1}{i+1}$$

and by the definition of supremum, we can choose $k_i > \max(K_i, k_{i-1})$ such that

$$a_{k_i} < p + \frac{1}{i+1}.$$

We have thus constructed a subsequence a_{n_k} such that $a_{n_k} \to p$. Hence $p \leqslant \limsup_{n\to\infty} a_n$, which is a contradiction.

In light of (1) and (2) above, we conclude that $\limsup_{n\to\infty} a_n = p$. □

Main Proof. Without loss of generality, suppose $\limsup_{n\to\infty} b_n$ is finite.

(1) If $\limsup_{n\to\infty} a_n = +\infty$, then

$$+\infty = \limsup_{n\to\infty}(a_n + b_n) = \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n = +\infty.$$

(2) If $\limsup_{n\to\infty} a_n = -\infty$, then

$$-\infty = \limsup_{n\to\infty}(a_n + b_n) = \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n = -\infty.$$

(3) If $\limsup_{n\to\infty} a_n$ is finite, then both a_n and b_n are bounded. Thus Theorem 3.6(b) of [**Rud76**] implies that $a_n + b_n$ has a convergent subsequence. Hence

$$\limsup_{n\to\infty}(a_n + b_n)$$

is finite. With the help of Lemma 3.1, we conclude that

$$\begin{aligned}
\limsup_{n\to\infty}(a_n + b_n) &= \lim_{k\to\infty} (\sup\{a_i + b_i : i \geqslant k\}) \\
&\leqslant \lim_{k\to\infty} (\sup\{a_i + b_j : i \geqslant k, j \geqslant k\}) \\
&= \lim_{k\to\infty} (\sup\{a_i : i \geqslant k\} + \sup\{b_j : j \geqslant k\}) \\
&= \lim_{k\to\infty} (\sup\{a_i : i \geqslant k\}) + \lim_{k\to\infty} (\sup\{b_j : j \geqslant k\}) \\
&= \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.
\end{aligned}$$

□

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